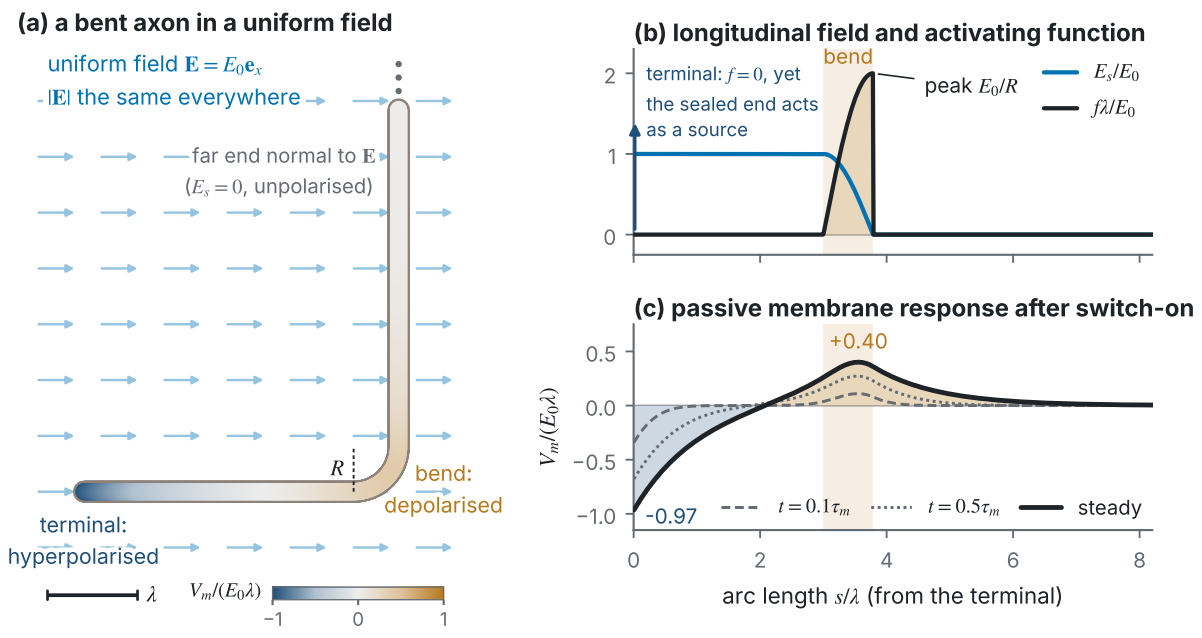

(a) a bent axon in a uniform field
uniform field $\mathbf{E} = E_0\mathbf{e}_x$
$|\mathbf{E}|$ the same everywhere
far end normal to $\mathbf{E}$
($E_s = 0$, unpolarised)
R
bend:
depolarised
terminal:
hyperpolarised
λ
$V_m/(E_0\lambda)$
−1
0
1
(b) longitudinal field and activating function
bend
terminal: $f = 0$, yet
the sealed end acts
as a source
peak E_0/R
E_s/E_0
$f\lambda/E_0$
2
1
0
(c) passive membrane response after switch-on
+0.40
-0.97
$t = 0.1\tau_m$
$t = 0.5\tau_m$
steady
$V_m/(E_0\lambda)$
0.5
0.0
−0.5
−1.0
0
2
4
6
8
arc length s/λ (from the terminal)